Angled and negative work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use force direction to determine work's sign and size.

SUCCESS CRITERIA - I can...

1. Compare force direction with displacement.
2. Use the parallel force component.
3. Explain positive, negative and zero work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A positive triangular pulse peaks at 50 N and lasts 0.3 s. It acts on a 3 kg cart initially at -2 m/s. Find impulse and final velocity.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $W = Fd\cos\theta = F_{parallel}\,d$.
θ is between force and displacement. CPA uses the supplied parallel component. Opposite force gives negative work.

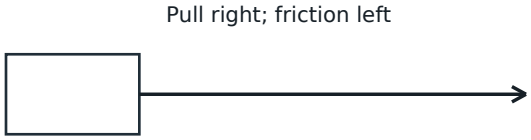
Worked reasoning: A box moves 4 m right while friction acts 3 N left. Friction's work is -3×4=-12 J because force opposes displacement.

Your turn: Draw force and displacement arrows. Add a perpendicular upward force and state its work for this horizontal motion.

Marzano check - circle: ① ② ③

03 Represent and interpret

The box moves right. Classify work by a rightward pull, leftward friction and upward normal force.



Motion right

The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

04 Solve and justify

A 20 N rope pulls 30° above horizontal over 5 m horizontally. Friction is 4 N. Find rope work, friction work and net work.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A force perpendicular to velocity can bend a path without changing speed. Explain using work.

Marzano check - circle: 1 2 3

06 Choose a method

Fragments of masses 4 and 6 kg separate from rest. The 4 kg piece moves at +12 m/s. Find the other velocity and total kinetic energy.

Marzano check - circle: 1 2 3

07 Exit check

A 10 N force acts at 120° to a 2 m displacement. Find work and explain the sign.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 20 N pull at 60° above horizontal moves a box 3 m right while friction is 4 N. Find work by pull, friction and their sum.

Marzano check - circle: 1 2 3